

Problem 7.4

It is known that an investment of \$500 will increase to \$4,000 at the end of 30 years. Find the sum of the present values of three payments of \$10,000 each which will occur at the end of 20, 40, and 60 years.

Solution

Let i be the effective annual interest rate. Then

$$4000 = 500(1 + i)^{30}.$$

Hence,

$$(1 + i)^{30} = 8,$$

and therefore

$$1 + i = 8^{1/30}, \quad i = 8^{1/30} - 1 \approx 0.071773,$$

so

$$i \approx 7.1773\%.$$

Also, the discount factor is

$$v = \frac{1}{1 + i} = 8^{-1/30}.$$

The sum of the present values is

$$10,000v^{20} + 10,000v^{40} + 10,000v^{60}.$$

Using $v = 8^{-1/30}$,

$$10,000v^{20} = 10,000 \cdot 8^{-20/30} \approx 2500.0215,$$

$$10,000v^{40} = 10,000 \cdot 8^{-40/30} \approx 625.0107,$$

$$10,000v^{60} = 10,000 \cdot 8^{-60/30} = 10,000 \cdot 8^{-2} = 156.2500.$$

Therefore,

$$2500.0215 + 625.0107 + 156.2500 = 3281.2822.$$

So the required sum of present values is

$$\boxed{\$3281.28}.$$